

EE 558

1. Snell's Law
 2. Total reflection, critical angle
 3. Planar waveguide, fiber waveguide.
 4. Spatial modes
 5. Single mode, multiple modes.
-

✓ Geometric optics : 

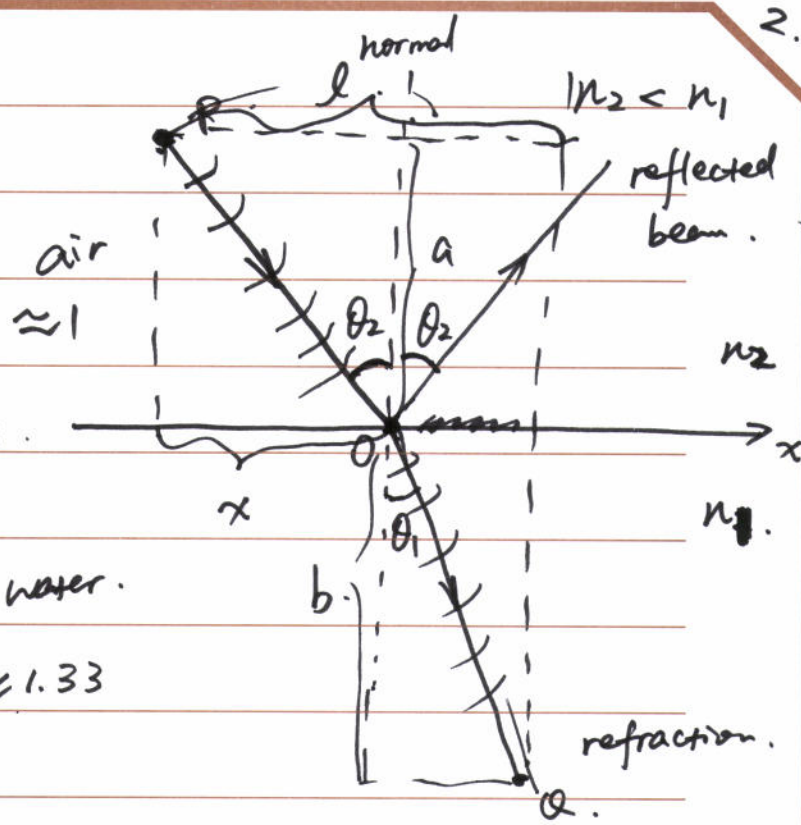
△ Wave optics : Maxwell's equations

1. Snell's Law.

$$v = \frac{c}{n}$$

$$\frac{\sin \theta_2}{\sin \theta_1} = \frac{v_2}{v_1} = \frac{n_1}{n_2}$$

$$\rightarrow n_2 \cdot \sin \theta_2 = n_1 \cdot \sin \theta_1$$



2.

Proof: Fermat principle.

$$T = t_{p,o} + t_{o,q} = \frac{n_2 \sqrt{x^2 + a^2}}{c} + \frac{n_1 \sqrt{b^2 + (l-x)^2}}{c}$$

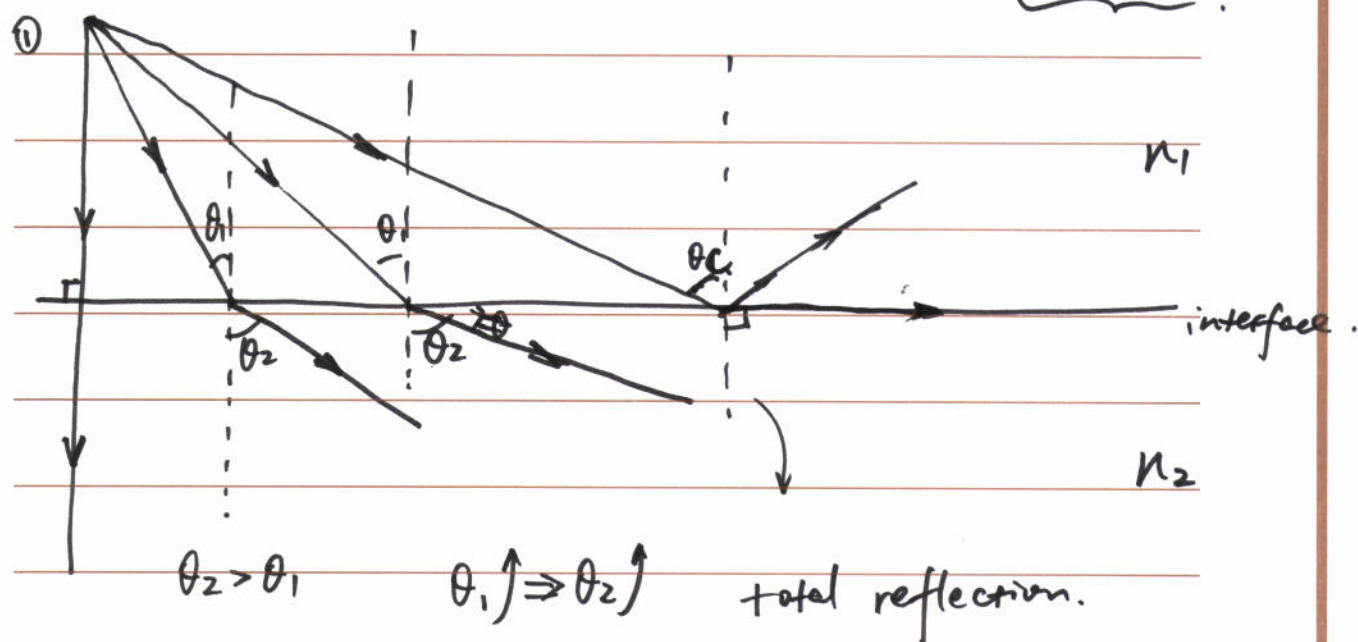
$$\frac{dT}{dx} = \frac{1}{2} \frac{n_2 \cdot 2x}{c \sqrt{x^2 + a^2}} + \frac{1}{2} \frac{n_1 \cdot 2(l-x) \cdot (-1)}{c \sqrt{b^2 + (l-x)^2}}$$

$$= \frac{n_2}{c} \cdot \frac{x}{\sqrt{x^2 + a^2}} - \frac{n_1}{c} \cdot \frac{l-x}{\sqrt{b^2 + (l-x)^2}}$$

$$= \frac{n_2}{c} \cdot \sin \theta_2 - \frac{n_1}{c} \cdot \sin \theta_1 = 0$$

2. Total internal reflection.

$$n_1 > n_2.$$



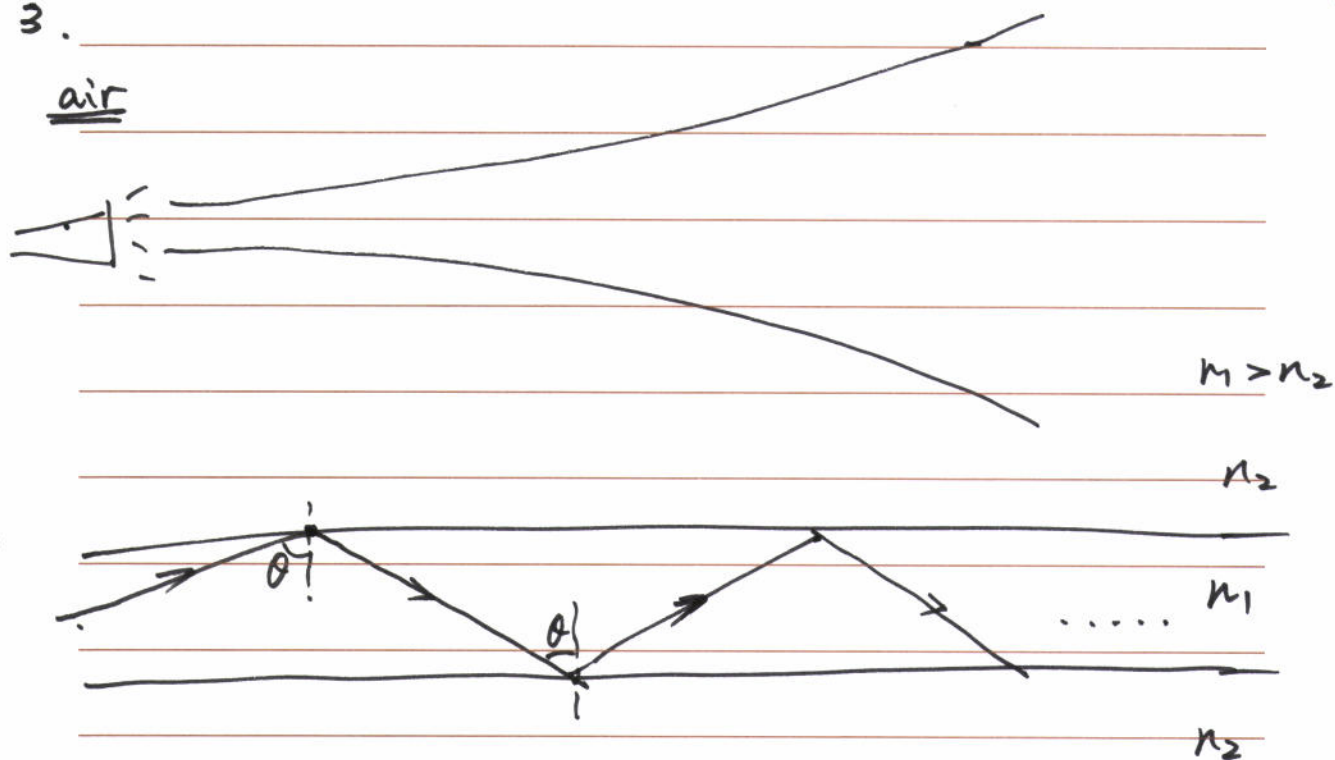
$$n_1 \cdot \sin \theta_1 = n_2 \cdot \sin \frac{\pi}{2}$$

$$\theta_c = \arcsin\left(\frac{n_2}{n_1}\right) \quad (n_2 < n_1).$$

↑
critical angle.

total reflection $\theta_1 \geq \theta_c$.

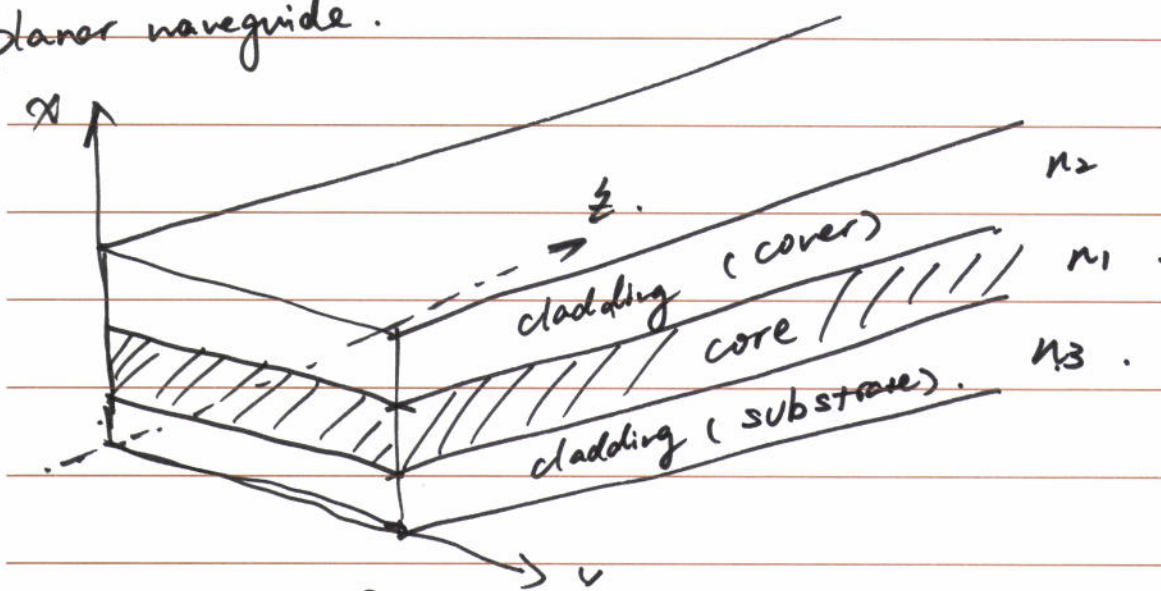
3.

air

if $\theta \geq \theta_c$

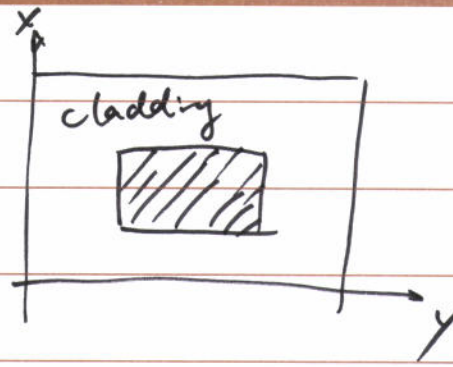
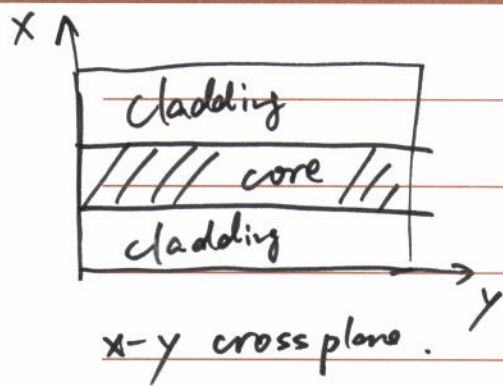
waveguide.

① planar waveguide.



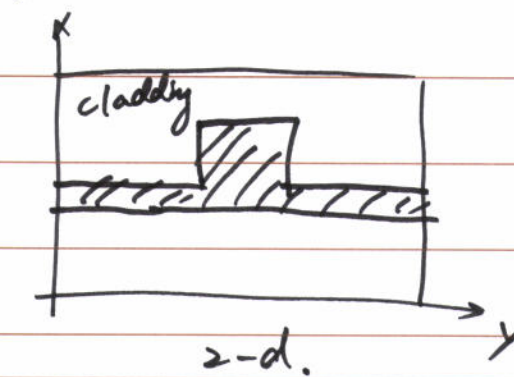
$n_1 > n_2$, & $n_1 > n_3$

$n_2 = n_3$ (Symmetric) $n_2 \neq n_3$ (non-symmetric).



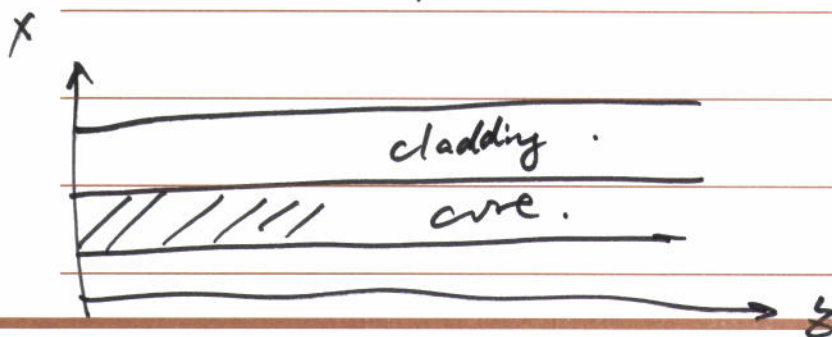
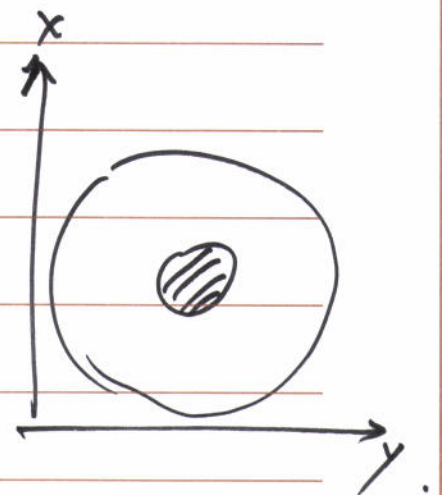
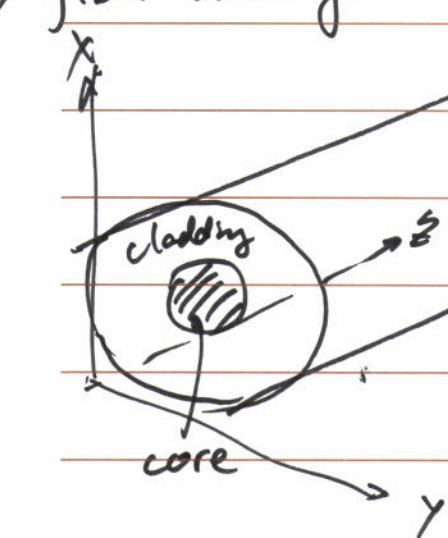
1-d.

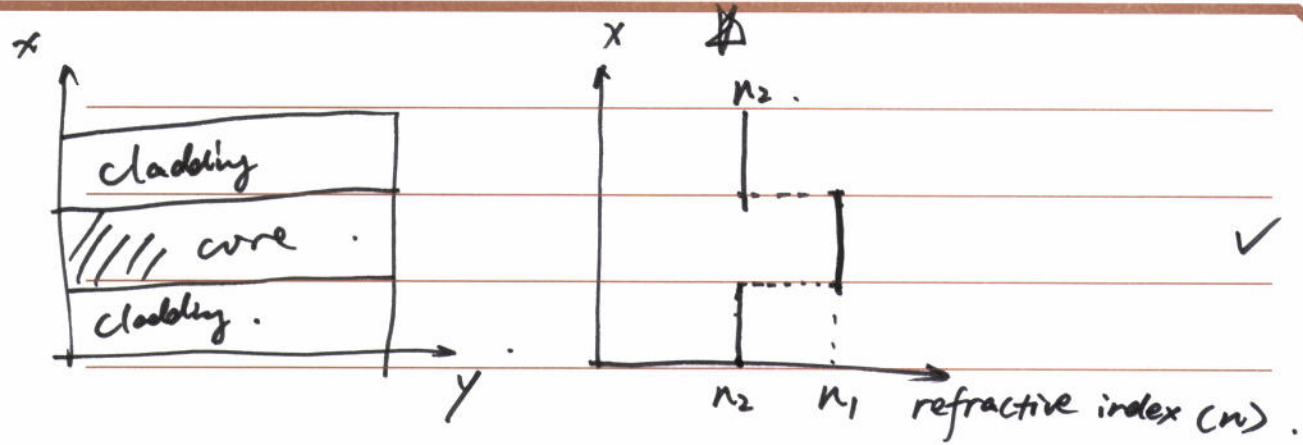
z-d.



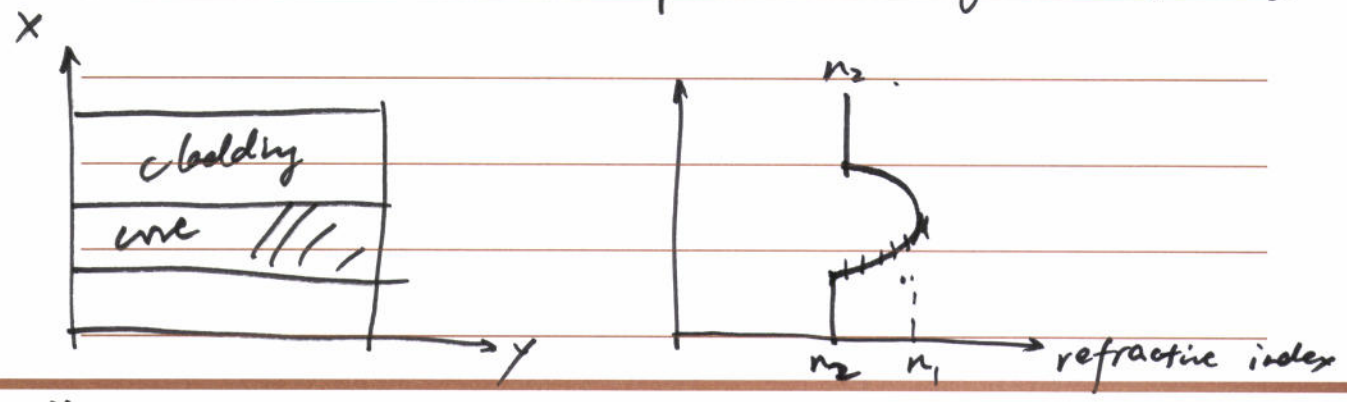
Ridge waveguide.

② fiber wave guide.

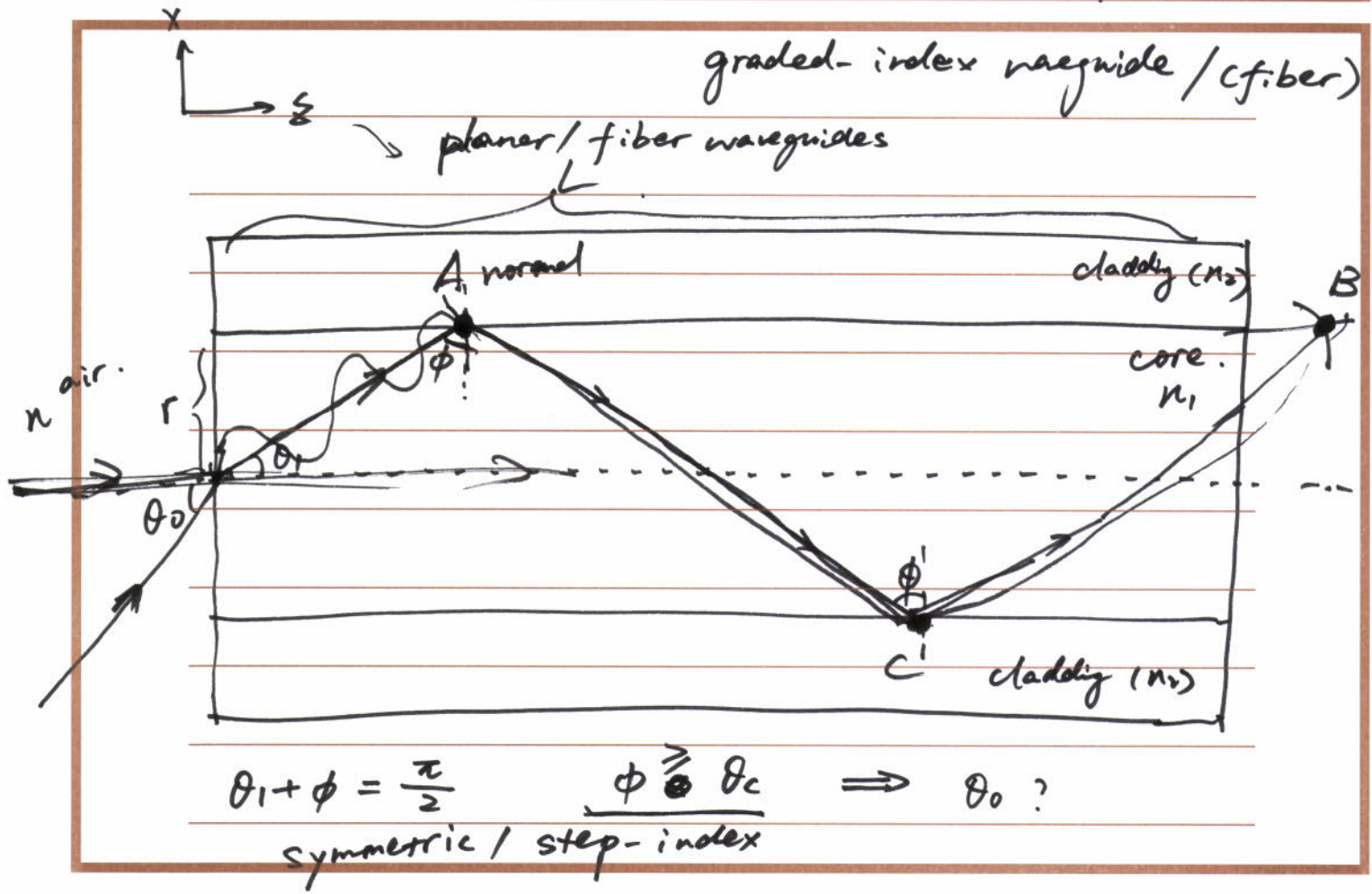




step-index waveguide / (fiber)



graded-index waveguide / (fiber)



$$\phi_{\min} = \arcsin\left(\frac{n_2}{n_1}\right)$$

$$n \cdot \sin \theta_{0, \max} = n_1 \sin \theta_1 = n_1 \cdot \cos \phi_{\min} = n_1 \cdot \frac{\sqrt{n_1^2 - n_2^2}}{n_1}$$

$$= \sqrt{n_1^2 - n_2^2} \quad (n_1 > n_2)$$

$$\text{if } n=1 : \sin \theta_{0, \max} = \sqrt{n_1^2 - n_2^2}$$

$$n \cdot \sin \theta_{0, \max} = N.A \quad (\text{Numerical Aperture})$$

$$n \cdot \sin \theta_{0, \max} \approx n_1 \sqrt{\frac{n_1^2 - n_2^2}{n_1^2}}$$

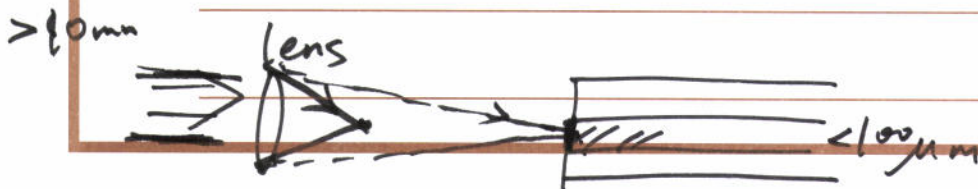
$$= n_1 \sqrt{\frac{(n_1 + n_2)(n_1 - n_2)}{n_1^2}}$$

$$n_1 \approx n_2, (n_1 > n_2) \quad \Delta = \frac{n_1 - n_2}{n_1} \quad 1\% \sim 5\%$$

$$N.A \approx n_1 \sqrt{2 \cdot \frac{n_1 - n_2}{n_1}} = n_1 \sqrt{2\Delta}$$

$$\begin{matrix} t_o \\ \wedge \\ t_i \end{matrix} \quad \Delta T = \frac{\left(\frac{L}{\sin \phi_c} - L\right)}{\frac{c}{n_1}} \approx \frac{L}{c} \cdot \frac{n_1^2}{n_2^2} \cdot \Delta$$

① $\theta_0 \rightarrow$ will be limited by the N.A



Q: Will all θ_0 sats. N.A. $\Rightarrow$ transmitted waveguide?

②. period. phase

stability.

$$\theta_A - \theta_B = 0, 2\pi, 4\pi, \dots, 2m\pi$$

★

$$2n_1 \cdot \frac{2\pi}{\lambda} \cdot 2r \cdot \cos\phi - \underbrace{\phi_A - \phi_C}_{\text{radius of core.}} = 2m\pi$$

↑
wavelength

↑
integer.

$m = 0, 1, 2, 3, \dots$ related waveguides $\left\{ \begin{array}{l} n_1/n_2 \\ \text{shape} \end{array} \right.$

basic mode.

for higher m , $\phi \downarrow$.

$\theta_0 \nearrow$ N.A. $\Rightarrow$ range

$\searrow$ ★ $\Rightarrow \theta_0$: discrete numbers.

for each $m(\phi) \rightarrow$ one spatial mode.

may have multiple possible modes to be transmitted.

① for planar waveguide step-indexed

$$V < \frac{\pi}{2} \leftarrow \text{single-mode waveguide.}$$

② for step-indexed fiber.

$$V < 2.41 \leftarrow \text{single-mode fiber}$$

② How many modes? (M)

if $M \leq 5$ — Maxwell's equations

else ($M > 5$)

approx. $\begin{cases} \text{shape} \\ n_1/n_2 \end{cases}$

① for planar waveguide. step-indexed.

$$M \approx \left[\frac{2V}{\pi} \right]$$

② for step-indexed fiber

$$M \approx \left[\frac{4V^2}{\pi^2} \right]$$

Example: fiber (stepped-index).

core: $n_1 = 1.5$, $\Delta = 0.01$, $\lambda = 1 \mu\text{m}$, $r = 30 \mu\text{m}$

cal: V , M ?

$$V = \frac{2\pi}{\lambda} \cdot r \cdot \sqrt{n_1^2 - n_2^2} \approx 40$$

$$M = \left[\frac{4V^2}{\pi^2} \right] = \underset{\Delta}{64}$$